Spare Part EMS 2.0 | Improvement Report: One platform to manage the full spare part lifecycle

TODAY'S PROCESS | Loses data, time and accuracy



Fragmented data

Excel and legacy run in parallel; 1-editor lock.

Missed purchase info

No bulk interlock/price check; high reject rate.



No cost control

No live cost or budget tracking; manual ETA chasing.



Wrong EBOM and WO

10 lines merged into 1 WO on wrong mechanism

EMS 2.0 SOLUTION | Six connected modules

0 ▪ Forecast & Stock

Predicts usage, triggers reorders, and caps inventory levels.

1 ▪ Mass Search

Price, interlock, photos, and 4 PO routes in one view.

2 ▪ Purchasing

Multi-user requests, automated routing, and audit trail.

3 ▪ Cost & ETA

Direct ETA tracking, over-budget reasons, cost booked on issue.

4 ▪ EBOM

Visual line and mechanism tree with automated sanity rules.

5 ▪ Work Orders

Mass WO creation with correct mechanism allocation.

Before and after at a glance

Area	Before (Legacy)	After (EMS 2.0)	Operational Impact
0 **Planning**	Reactive; no forecast or safety stock	Auto forecast, reorder alerts, stock cap	**Proactive planning**
1 **Search**	Excel plus legacy system; no bulk lookup	Bulk lookup of price, interlock, photos, 4 PO routes	**−80% lookup time**
2 **Purchasing**	One editor per file; rejects get missed	Real-time multi-user; bulk handling; audit log	**No file locks, no lost orders**
3 **ETA & cost**	ETA via intermediaries; no live cost view	Direct ETA updates; overdue alerts; cost at issue	**Real-time budget control**
4 **EBOM**	Hard to read; missing groups unnoticed	Visual tree; sanity rules; fast cloning	**Standardized EBOM data**
5 **Work Orders**	Manual; 1 WO covers 10 lines	Bulk WO split by line and mechanism	**Accurate MTBF / MTTR**

Modules 0–1: Plan ahead, check before ordering

0 ▪ Forecast & Safety Stock

BEFORE

- No usage forecast; ordering is reactive
- Long-lead parts are ordered too late
- No safety stock: over- or under-buying

AFTER

- Auto forecast from production output and time
- Automated reorder alerts when an order is due
- Safety-stock limit configured per part
- Bulk Q-Code onboarding and registration

1 ▪ Mass Search

BEFORE

- Excel and legacy system checked separately
- One item at a time; no bulk lookup capability
- Orders rejected downstream after registration

AFTER

- Bulk lookup: price, interlock, stock, photos
- Status of all 4 PO routes visible in one view
- Critical part issues flagged before ordering

Module 2: Four PO routes, auto-escalated

STEP 1
Stock transfer
Use entity stock first.
Locked / rejected → Step 2

→

STEP 2
Internal PO
Buy via entity if cheaper.
Rejected → Step 3

→

STEP 3
Standard PO
Regular order with SLA and ETA tracking.

→

STEP 4
Urgent PO
Fast approval for line downtime needs.

BEFORE

- One Excel editor at a time (U-dock file lock)
- Multi-layer review: LEC, Reviewer, Requestor
- Rejects rechecked by hand; information gets lost
- ETA chased manually through intermediaries

AFTER

- Real-time multi-user work; replaces U-dock
- LEC handles hundreds of items in bulk
- In-system comments with an immutable audit trail
- Overdue orders highlighted; ETA updated in-system

Module 3: Cost counted when parts are used

TODAY

- No real-time cost or over-budget tracking
- Cost not tied to physical issue date
- Orders disconnected from stock and budget
- No view of issue-plan status per item

1 ▪ Cost on consumption

Cost is booked only when a part is issued, ensuring each period's budget is accurate.

2 ▪ ETA & scheduling

Track arrival dates and schedule orders smoothly without cost spikes.

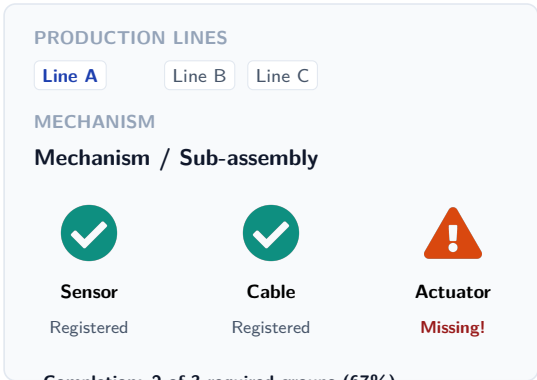
3 ▪ Over-budget control

Live budget tracking. A mandatory reason code is required for every overrun.

4 ▪ Issue-plan tracking

Every item is clearly classified as planned, issued, or unissued.

Module 4: EBOM you can see and trust



TODAY

- No overview across parallel lines and mechanisms
- Wrong or missing part groups go completely unnoticed
- No bulk registration or visual progress tracking

WHAT CHANGES

- Visual view of all lines and mechanism trees
- Sanity rules flag missing groups (Sensor, Cable, Actuator)
- Mass cloning across identical lines in a single step
- Real-time completion % per mechanism and line

Module 5: One action, every work order

BEFORE

10 lines use the same part

1 WO ⟶ 1 mechanism

Line 1 takes 10× cost; other 9 lines show zero cost.

AFTER

10 lines use the same part

10 WOs, each on its own mechanism

100% accurate cost allocation per line and mechanism.

Less manual work

A single action automatically generates and releases all work orders in bulk.

Correct allocation

Parts post accurately to the specific mechanism of the exact line consuming them.

Reliable history

Issuance records reflect shop floor reality for precise MTBF/MTTR audits.

Impact: faster, more accurate, fully traceable

01 ▪ SPEED

No more Excel file locks

Mass Search, Request, EBOM and WO cut manual operations by **60–70%**.

02 ▪ ACCURACY

Standardized EBOM & WO

Interlock and pricing verified prior to ordering. WOs land on the correct mechanism.

03 ▪ TRACEABILITY

Live 4-route PO & SLA

Collaborative comments with full audit log. Overdue orders highlighted automatically.

04 ▪ COST & STOCK

Cost counted at actual use

Predictive forecast and safety stock limits. Over-budget root cause codes enforced.